

A PROJECT REPORT

on

AI RESUME BUILDER

Submitted by

Ms. SAKSHI VASANT SHINDE

in partial fulfillment for the award of the degree

of

BACHELOR OF SCIENCE

In

COMPUTER SCIENCE

under the guidance of

Mrs. Manisha Firake

Department of Computer Science



**MAHATMA GANDHI MISSION'S
COLLEGE OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY**

Sector-1, Kamothe, Navi Mumbai 410209

(Sem VI)

(2022-2025)



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CERTIFICATE

This is to certify that Ms. Sakshi Vasant Shinde of T.Y.B.Sc. (Sem V) class Seat No. 3008244 satisfactorily completed the Project AI Resume Builder, to be submitted in the partial fulfillment for the award of Bachelor of Science in Computer Science during the academic year 2022 – 2025.

Date of Submission:

Project Guide

Head/In charge

Department of Computer Science

College Seal

Signature of Examiner

DECLARATION

I, **Sakshi Vasant Shinde**, hereby declare that the project entitled **“AI Resume Builder”** submitted in the partial fulfillment for the award of Bachelor of Science in Computer Science during the academic year **2022 – 2025** is my original work and the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

Signature of the Student:

Place:

Date:

Abstract

AI Resume Builder

The **AI Resume Builder** is a web-based application developed to help users create professional resumes quickly and efficiently using Artificial Intelligence. The system streamlines the resume creation process by allowing users to enter their personal, educational, and professional details through a structured form. Based on the provided input, the AI suggests optimized summaries and section content relevant to the user's profile.

The application includes essential features such as user authentication, a single pre-designed resume template, real-time resume preview, and PDF download functionality. The system is built using modern technologies such as ReactJS for the frontend, TailwindCSS for styling, Clerk for user authentication, and Strapi as the backend content management system.

This project aims to reduce the effort required to create an effective and professional resume. It is particularly beneficial for students and job seekers who need guidance in presenting their information in a well-structured and industry-standard format.

Acknowledgement

Ms. Sakshi Vasant Shinde the final year student of Bachelor of Science in Computer Science from MGM College of Computer Science and Information Technology would like to express my heartfelt gratitude to all those who have supported and inspired me throughout this journey. A special thanks to my mentor and HOD Mrs. Manisha Firake for his continuous guidance which shaped by understanding.

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Chapter 1

Introduction

1.1 Background

In the current job market, a resume acts as the first impression for any candidate. A well-structured, professional resume plays a crucial role in securing job opportunities. However, many students and job seekers struggle with formatting and writing content that aligns with industry expectations. This leads to missed opportunities and weak first impressions.

With the advancement of Artificial Intelligence, it is now possible to automate and personalize resume creation. The AI Resume Builder project was developed with the intention of assisting users in generating high-quality resumes with minimal effort. It uses intelligent suggestions for summaries and other resume sections, making it especially useful for freshers and non-technical users.

1.2 Objectives

The main objective of the AI Resume Builder project is to create a smart, user-friendly web application that helps users—especially students and freshers—easily build professional resumes. By integrating Artificial Intelligence, the system offers real-time content suggestions to ensure the final resume is well-structured, optimized, and job-ready.

1. Streamline Resume Creation
2. Enhance User Experience
3. Provide Real-Time Content Suggestions
4. Efficient Resume Management
5. Ensure Data Security and Privacy

1.3 Purpose and Scope

Purpose

The purpose of this project is to create a digital tool that simplifies the resume-building process using AI-powered suggestions. It is aimed at helping users, especially fresh graduates and job seekers, to create resumes that match current industry standards.

Scope

1. Resume Content Suggestions:
AI-driven suggestions for career objectives, summaries, skills, and other resume sections.
2. Resume Creation
Users can input their personal, educational, and professional details to generate a customized resume.
3. Real-Time Preview
Live preview of the resume with real-time updates as users input their information.
4. User Accounts
Users can create and manage accounts to save, edit, and download their resumes.
5. Support Services
A help section with FAQs, contact details, and guidelines on creating effective resumes.

1.4 Target Audience

This web application caters to a diverse audience, including:

- 1. Students and Freshers**

Those applying for internships or entry-level job roles.

- 2. Job Seekers**

Individuals looking for quick and professional resume-building solutions.

- 3. Non-technical Users**

Users who lack experience in formatting or writing resumes.

Chapter 2

System Analysis

2.1 Existing System

Many online platforms are available for resume building, each serving specific purposes. However, there is often a gap in user interaction and personalization. Most existing systems lack real-time guidance and intelligent support, making it difficult for users—especially freshers—to create effective resumes. Since these platforms handle a large amount of user data, security and privacy also become major concerns.

2.2 Proposed System

The proposed system is an AI-powered resume builder designed to help users create professional resumes with ease. It provides smart suggestions for content based on user input, job role, and industry standards. The system includes features like resume templates, real-time editing, and customization options. It also ensures the security of users' personal information, such as login credentials and resume data. This intelligent and user-friendly platform aims to bridge the gap between users and technology by offering a guided, interactive resume-building experience.

2.3 Requirement Analysis

The requirement analysis lays the foundation for developing a reliable and intelligent AI Resume Builder that caters to user needs while remaining scalable, secure, and easy to use. This phase involves understanding the expectations of job seekers, especially freshers, aligning with industry trends, and defining the technical and functional requirements. The goal is to ensure that the system provides an interactive, user-friendly, and personalized resume-building experience powered by AI, while also protecting user data and maintaining high performance.

2.4 Hardware Requirements

Processor	Intel Core i5 / Ryzen 5 or higher
RAM	Minimum 8 GB
Storage	Minimum 256 GB SSD
Display	13” or larger (recommended)
Internet Connection	Stable internet connection

2.5 Software Requirements

OS	Windows 10 / 11 or Linux
Code Editor	Visual Studio Code
Frontend	• React JS with Vite • Tailwind CSS
Backend	Strapi
Database	MySQL

2.6 Justification of Selection of Technology

The AI Resume Builder application is designed to provide seamless performance for users and ensure smooth operations. The frontend, built with React and styled using Tailwind CSS, enhances user interactivity and offers an excellent user experience. The backend utilizes Strapi, a flexible headless CMS, while Clerk is used for secure authentication. Data is managed efficiently with a relational database, such as MySQL. These technologies work together to ensure a robust, user-friendly, and secure web application.

2.7 Planning

Task Name	Duration	Start	Finish
Project topic inception	5 days	20/1/25	24/1/25
Preparing SRS	4 days	25/1/25	28/1/25
Requirement analysis	10 days	29/1/25	7/2/25
UML diagrams	3 days	8/2/25	10/2/25
Feasibility study	3 days	11/2/25	13/2/25
System design and planning	10 days	14/2/25	23/2/25
Coding and implementation	20 days	24/2/25	15/3/25
Testing	3 days	16/3/25	18/3/25
Integration/Validation	2 days	19/3/25	20/3/25
System testing	5 days	21/3/25	25/3/25

Chapter 3

System Design

3.1 Module Division

The system is divided into distinct modules, each focusing on specific features like user authentication, resume creation, and AI suggestions. This allows each part to function independently, making the application easier to develop and scale.

1. User Authentication Module

- **Purpose:**
 - Secure user login, registration, and profile management.
- **Features:**
 - User registration and login.
 - Password reset and email verification.
 - Login by multiple platforms like LinkedIn, GitHub.
- **Technologies:**
 - Clerk for authentication and user management.

2. Resume Creation Module

- **Purpose:**
 - Allow users to create, edit, and save resumes easily.
- **Features:**
 - Real-time text editing and customization.
 - Sections for personal information, skills, experience, and education.
- **Technologies:**
 - **React** for frontend framework.
 - **Tailwind CSS** for responsive design and styling.

3. AI Suggestions Module

- **Purpose:**
 - Provide personalized suggestions based on the user's input.
- **Features:**
 - Suggest keywords, and industry-related phrases.
 - Automatically optimize content.
- **Technologies:**
 - Gemini API Model

4. Resume Preview and Download Module

- **Purpose:**
 - Allow users to preview and download their completed resumes.
- **Features:**
 - Live preview of the resume.
 - Option to download as PDF.
 - Option to save the resume.
- **Technologies:**
 - **React** for rendering live preview and handling the download functionality.

5. User Profile Management Module

- **Purpose:**
 - Manage user information and maintain previous resume versions.
- **Features:**
 - View and edit personal information (name, contact details).
 - View history of created resumes.
- **Technologies:**
 - Strapi for backend content management and storing user data.

6. Admin Dashboard Module

- **Purpose:**
 - Provide administrators with tools to manage users and monitor platform usage.
- **Features:**
 - Manage user accounts and permissions.
 - View analytics on platform usage and user activity.
- **Technologies:**
 - React for the frontend interface.
 - Strapi for content management and analytics.

7. Security and Data Protection Module

- **Purpose:**
 - Ensure the safety of user data and protect privacy.
- **Features:**
 - Data encryption for sensitive information (e.g., passwords, resumes).
 - Secure data transmission using HTTPS and JWT for authentication.
- **Technologies:**
 - Clerk for secure authentication.
 - HTTPS for secure data handling.

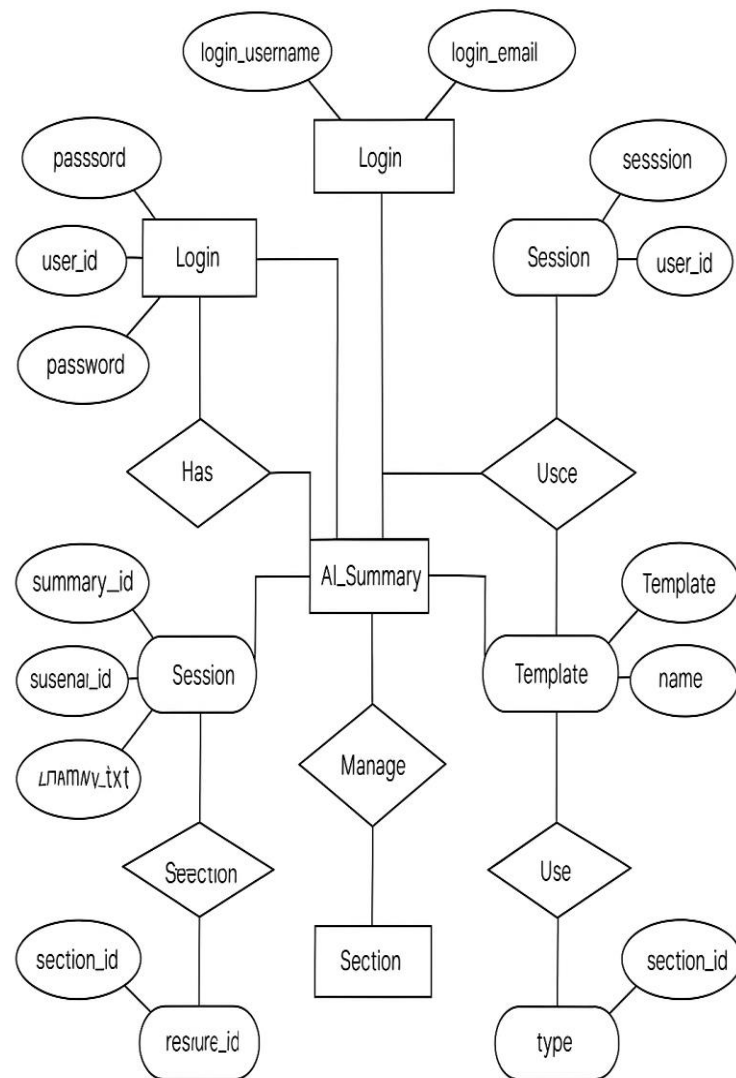
8. Database Module

- **Purpose:**
 - Store and manage user and resume data securely.
- **Features:**
 - Store user profiles, resume data, and metadata.
 - Maintain database integrity and backup.
- **Technologies:**
 - **MySQL** for relational database management with Xampp hosting.

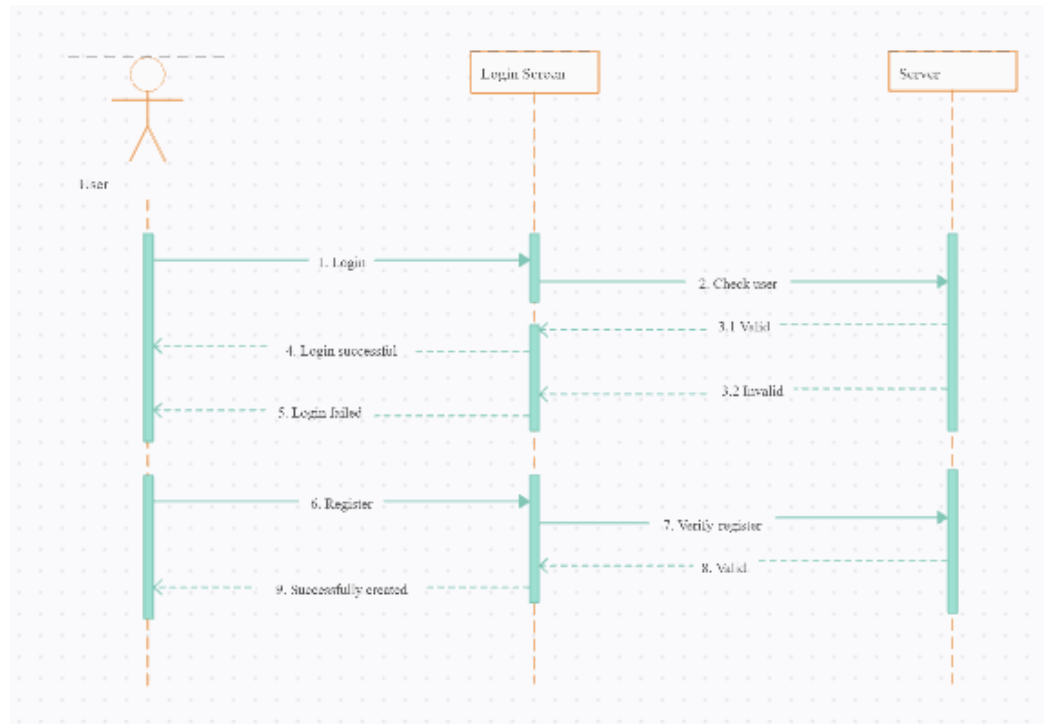
3.2 Data Dictionary

- 1.user_id: Unique identifier for a user.
- 2.username: Username of the registered user.
- 3.email: Email address of the user.
- 4.password: Encrypted password of the user.
- 5.resumeId: Unique identifier for the resume.
- 6.title: Title or name of the resume.
- 7.summary: Short professional summary.
- 8.experienceId: Unique identifier for the experience entry.
- 9.jobTitle: Job title held in a specific experience entry.
- 10.companyName: Name of the company for the experience.
- 11.city: Name of city for experience.
- 12.state: Name of the state for experience.
- 13.startDate: Start date of employment.
- 14.endDate: End date of employment.
- 15.description: Responsibilities or achievements in the experience.
- 16.university: Name of the university for the education.
- 17.degree: Degree obtained during education.
- 18.major: Major or specialization in the education.
- 19.startDate: Start date of education.
- 20.endDate: End date of education.
- 21.skillId: Unique identifier for the skill.
- 22.skill: Name of the skill.

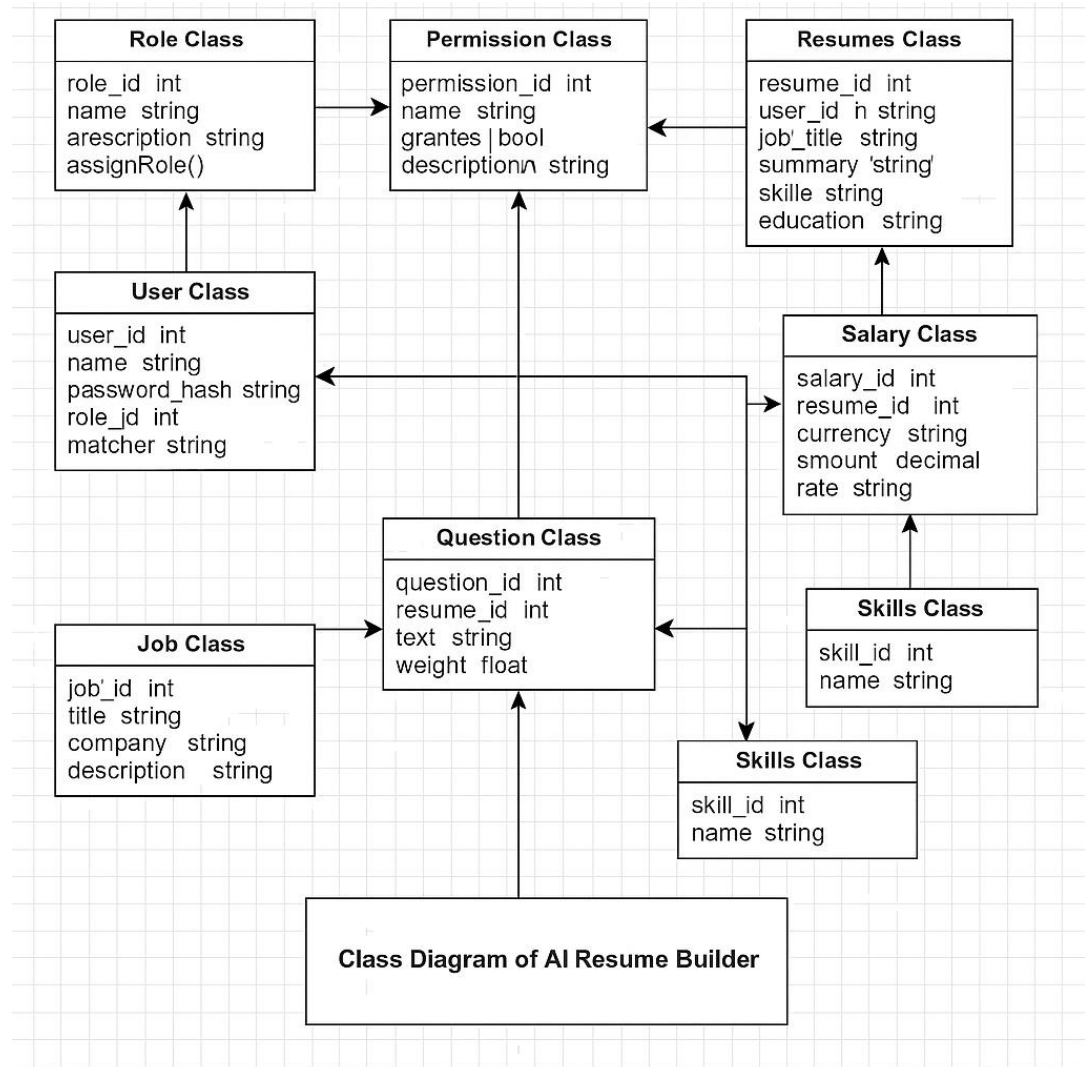
3.3 ER Diagram



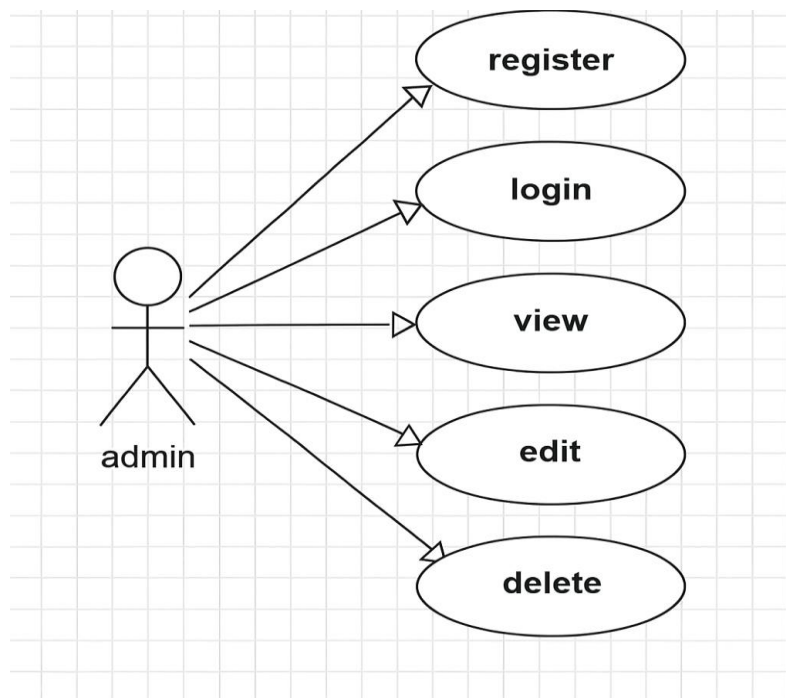
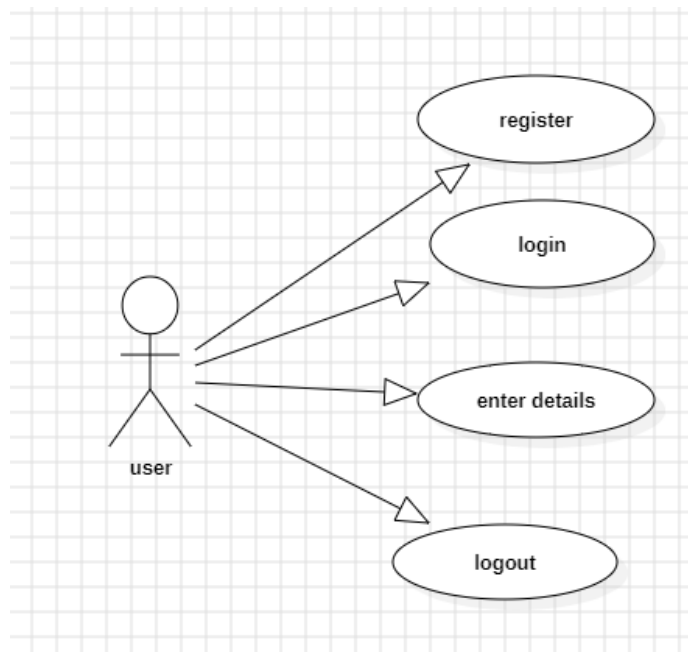
3.4 Sequence



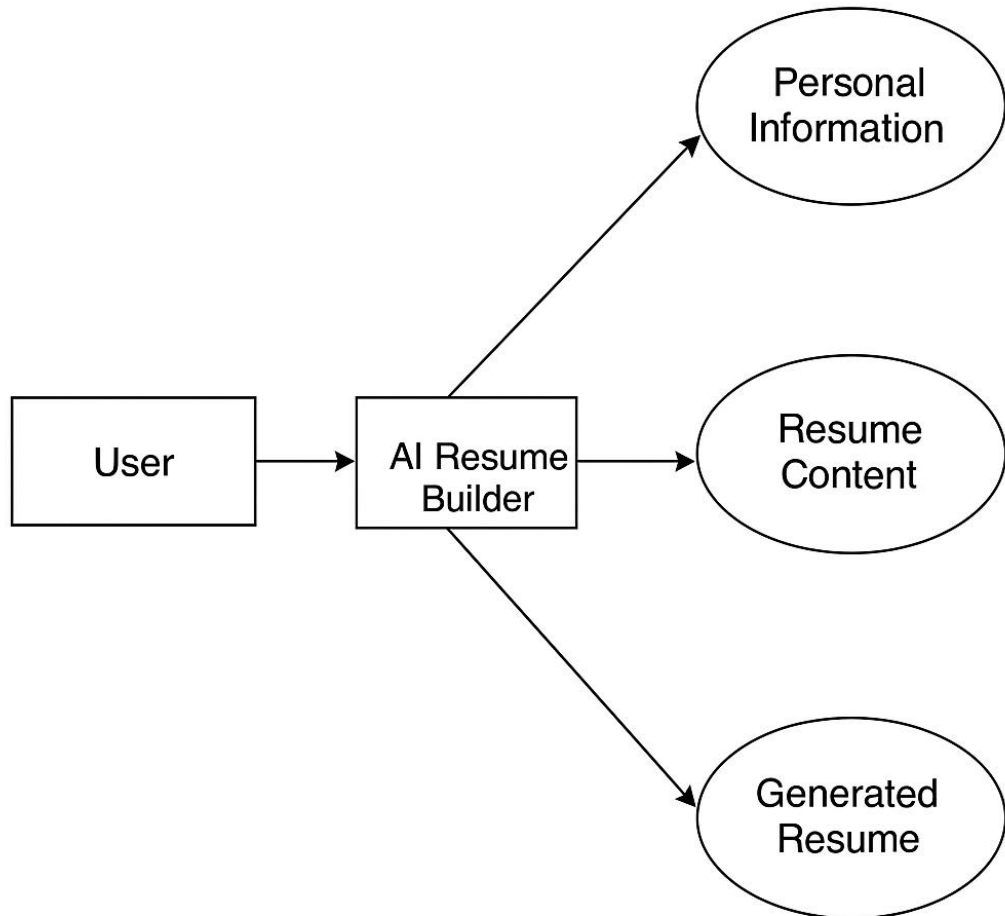
3.5 Class Diagram

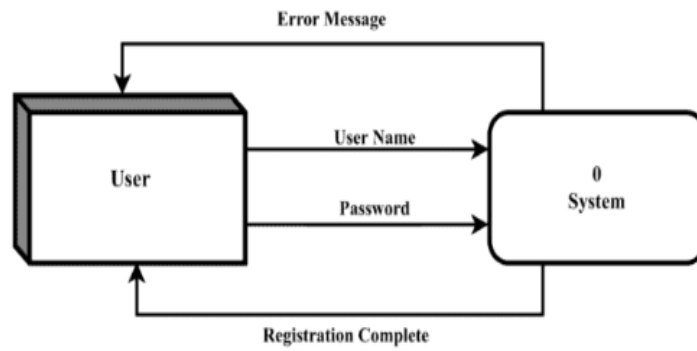


3.6 Use Case Diagram

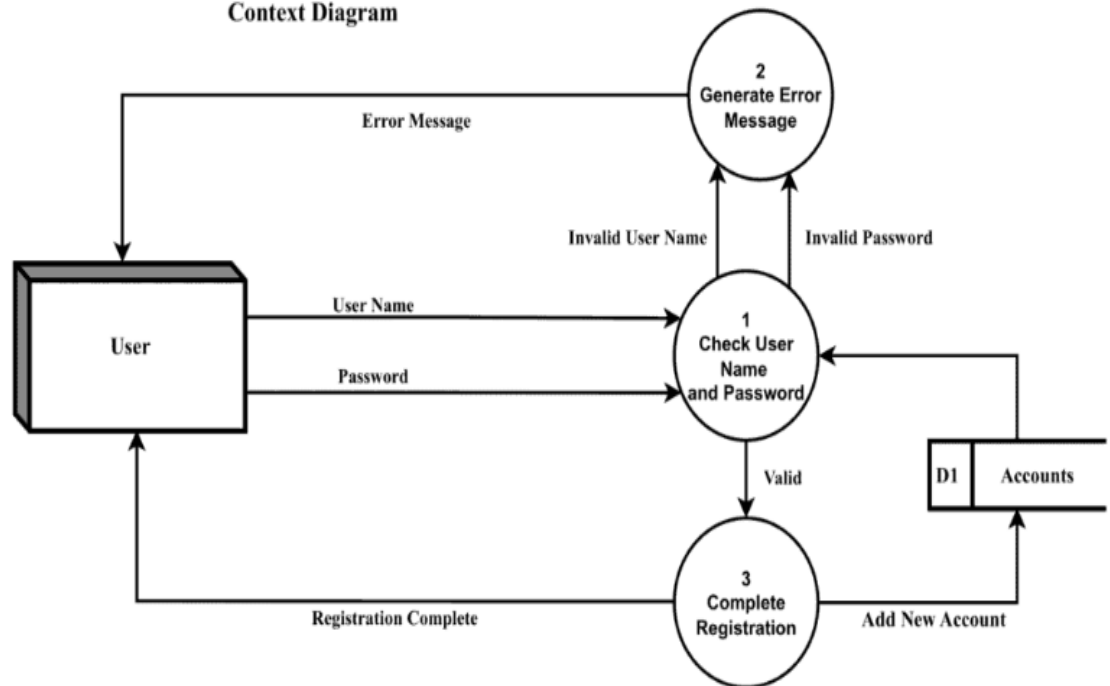


3.7 Data Flow Diagram



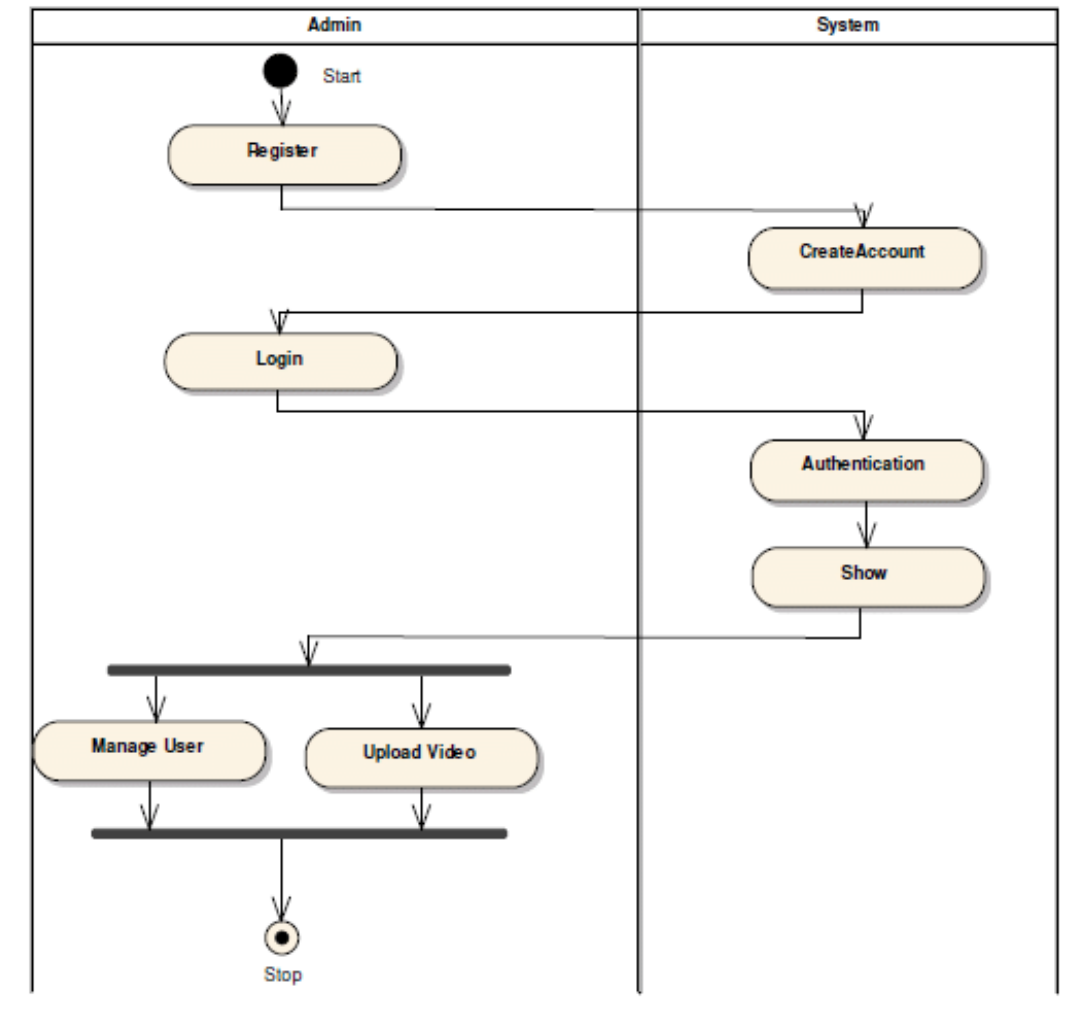


Context Diagram



3.8 Activity Diagram

ct Activity Diagram



Chapter 4

Implementation and Testing

4.1 Code

Home/index.jsx

```
import Header from '@components/custom/Header'
import { UserButton } from '@clerk/clerk-react'
import { AtomIcon, Edit, Share2 } from 'lucide-react'
import React from 'react'

function Home() {
  return (
    <div>
      <Header/>
      <div>
        <section className=" z-50">
          <div className="py-8 px-4 mx-auto max-w-screen-xl text-center lg:py-16 lg:px-12">
            <h1 className="mb-4 text-4xl font-extrabold tracking-tight leading-none text-gray-900 md:text-5xl lg:text-6xl dark:text-white">
              Build Your Resume <span className='text-primary'>With AI</span> </h1>
              <p className="mb-8 text-lg font-normal text-gray-500 lg:text-xl sm:px-16 xl:px-48 dark:text-gray-400">Effortlessly Craft a Standout Resume with Our AI-Powered Builder</p>
              <div className="flex flex-col mb-8 lg:mb-16 space-y-4 sm:flex-row sm:justify-center sm:space-y-0 sm:space-x-4">
                <a href="/dashboard" className="inline-flex justify-center items-center py-3 px-5 text-base font-medium text-center text-white rounded-lg bg-primary hover:bg-primary focus:ring-4 focus:ring-primary-300 dark:focus:ring-primary-900">
                  Get Started
                </a>
              </div>
            </div>
          </section>
        </div>
      </div>
    </div>
  )
}
```

```
      <svg className="ml-2 -mr-1 w-5 h-5" fill="currentColor" viewBox="0 0
20 20" xmlns="http://www.w3.org/2000/svg"><path fill-rule="evenodd" d="M10.293
3.293a1 1 0 011.414 0l6 6a1 1 0 010 1.414l-6 6a1 1 0 01-1.414-1.414L14.586 11H3a1
1 0 110-2h11.586l-4.293-4.293a1 1 0 010-1.414z" clip-
rule="evenodd"></path></svg>
```

```
    </a>
```

```
  </div>
```

```
</div>
```

```
</section>
```

```
<section className="py-8 bg-white z-50 px-4 mx-auto max-w-screen-xl text-center
lg:py-16 lg:px-12">
```

```
<h2 className="font-bold text-3xl">How it Works?</h2>
```

```
<h2 className="text-md text-gray-500">Craft your perfect resume in just a few
clicks — no hassle, just results.</h2>
```

```
<div className="mt-8 grid grid-cols-1 gap-8 md:grid-cols-2 lg:grid-cols-3">
```

```
  <a
```

```
    className="block rounded-xl border bg-white
```

```
    border-gray-200 p-8 shadow-xl transition
```

```
    hover:border-pink-500/10 hover:shadow-pink-500/10"
```

```
    href="#"
```

```
  >
```

```
    <AtomIcon className="h-8 w-8"/>
```

```
    <h2 className="mt-4 text-xl font-bold text-black">Write Your Resume
Prompt</h2>
```

```
    <p className="mt-1 text-sm text-gray-600">
```

```
      Just tell us about your education, experience, and skills
```

```
    </p>
```

```
  </a>
```

```
  <a
```

```
    className="block rounded-xl border bg-white border-gray-200 p-8 shadow-xl
transition hover:border-pink-500/10 hover:shadow-pink-500/10"
```

```

      href="#"
    >
    <Edit className='h-8 w-8' />
    <h2 className="mt-4 text-xl font-bold text-black"> Customize & Edit Easily
  </h2>
    <p className="mt-1 text-sm text-gray-600">
      Make changes to the design, content, and layout in a simple, drag-and-drop
      editor.
    </p>
    </a>
    <a
      className="block rounded-xl border bg-white border-gray-200 p-8 shadow-xl
      transition hover:border-pink-500/10 hover:shadow-pink-500/10"
      href="#"
    >
    <Share2 className='h-8 w-8' />
    <h2 className="mt-4 text-xl font-bold text-black">Download & Share
    Instantly</h2>
    <p className="mt-1 text-sm text-gray-600">
      Grab your professionally crafted resume in seconds.
    </p>
    </a>
  </div>
</section>
</div>
</div>
)
}
export default Home

```

auth/sign-in/index.jsx

```
import { SignIn } from '@clerk/clerk-react'

import React from 'react'

function SignInPage() {
  return (
    <div className='flex justify-center my-20 items-center'>
      <SignIn/>
    </div>
  )
}

export default SignInPage
```

Dashboard/index.jsx

```
import React, { useEffect, useState } from 'react'
import AddResume from '../components/AddResume'
import { useUser } from '@clerk/clerk-react'
import GlobalApi from '../service/GlobalApi';
import ResumeCardItem from '../components/ResumeCardItem';

function Dashboard() {
  const {user}=useUser();
  const [resumeList,setResumeList]=useState([]);
  useEffect(()=>{
    user&&GetResumesList()
  },[user])
  /**
   * Used to Get Users Resume List
   */
  const GetResumesList=()=>{
    GlobalApi.GetUserResumes(user?.primaryEmailAddress?.emailAddress)
```

```

.then(resp=>{
  console.log(resp.data.data);
  setResumeList(resp.data.data);
})
}
return (
  <div className='p-10 md:px-20 lg:px-32'>
    <h2 className='font-bold text-3xl'>My Resume</h2>
    <p>Start creating your resume with AI Resume Builder</p>
    <div className='grid grid-cols-2
md:grid-cols-3 lg:grid-cols-5 gap-4
mt-10
'>
      <AddResume/>
      {resumeList.length>0?resumeList.map((resume,index)=>(
        <ResumeCardItem resume={resume} key={index}
refreshData={GetResumesList}/>
      )):
      [1,2,3,4].map((item,index)=>(
        <div className='h-[280px] rounded-lg bg-slate-200 animate-pulse'>
        </div>
      ))
    }
  </div>
</div>
)
}
export default Dashboard

```

database.js

```
const path = require('path');

module.exports = ({ env }) => {

  const client = env('DATABASE_CLIENT', 'sqlite');

  const connections = {

    mysql: {
      connection: {
        host: env('DATABASE_HOST', 'localhost'),
        port: env.int('DATABASE_PORT', 3306),
        database: env('DATABASE_NAME', 'ai-resume-builder'),
        user: env('DATABASE_USERNAME', 'root'),
        password: env('DATABASE_PASSWORD', ''),
        ssl: env.bool('DATABASE_SSL', false) && {
          key: env('DATABASE_SSL_KEY', undefined),
          cert: env('DATABASE_SSL_CERT', undefined),
          ca: env('DATABASE_SSL_CA', undefined),
          capath: env('DATABASE_SSL_CAPATH', undefined),
          cipher: env('DATABASE_SSL_CIPHER', undefined),
          rejectUnauthorized:
env.bool('DATABASE_SSL_REJECT_UNAUTHORIZED', true),
        },
      },
      pool: { min: env.int('DATABASE_POOL_MIN', 2), max:
env.int('DATABASE_POOL_MAX', 10) },
    },

    postgres: {
      connection: {
        connectionString: env('DATABASE_URL'),
        host: env('DATABASE_HOST', 'localhost'),
        port: env.int('DATABASE_PORT', 5432),
```

```

    database: env('DATABASE_NAME', 'strapi'),
    user: env('DATABASE_USERNAME', 'strapi'),
    password: env('DATABASE_PASSWORD', 'strapi'),
    ssl: env.bool('DATABASE_SSL', false) && {
      key: env('DATABASE_SSL_KEY', undefined),
      cert: env('DATABASE_SSL_CERT', undefined),
      ca: env('DATABASE_SSL_CA', undefined),
      capath: env('DATABASE_SSL_CAPATH', undefined),
      cipher: env('DATABASE_SSL_CIPHER', undefined),
      rejectUnauthorized:
env.bool('DATABASE_SSL_REJECT_UNAUTHORIZED', true),
    },
    schema: env('DATABASE_SCHEMA', 'public'),
  },
  pool: { min: env.int('DATABASE_POOL_MIN', 2), max:
env.int('DATABASE_POOL_MAX', 10) },
},
  sqlite: {
    connection: {
      filename: path.join(__dirname, '..', env('DATABASE_FILENAME',
'.tmp/data.db')),
    },
    useNullAsDefault: true,
  },
};
return {
  connection: {
    client,
    ...connections[client],

```

```
        acquireConnectionTimeout: env.int('DATABASE_CONNECTION_TIMEOUT',  
60000),  
    },  
};  
};
```

4.2 Testing Approach

1. Unit Testing

Unit testing verifies the smallest testable parts of the application independently to ensure they function as expected.

- **Tools:** Jest (React components), unit test(backend)
- **What to Test:**
 - Form input validation (email, password, job title, etc.)
 - AI summary generator function
 - Resume component rendering
 - API call handlers (submit, edit, delete resume sections)

2. Integration Testing

Integration testing checks the interaction between multiple components/modules in the system.

- **Tools:** Postman, Supertest (Node.js)
- **What to Test:**
 - User signup/login → dashboard access
 - Resume creation → add experience → download
 - AI input → output → save to database
 - API communication between frontend and backend

3. End-to-End (E2E) Testing

E2E testing simulates real user scenarios to ensure the system works from start to finish.

- **Tools:** Cypress, Selenium, Playwright
- **What to Test:**
 - New user registration and login
 - Creating, editing, and downloading resumes
 - Generating AI summary from job description
 - Logging out and logging back in

4. Performance Testing

Performance testing checks how well the AI Resume Builder performs under different conditions such as high user load, AI usage peaks, and resume generation stress. The goal is to ensure that the application is responsive and scalable.

- **Tools:** JMeter, Locust, Apache Benchmark, Lighthouse (frontend)
- **What to Test:**
 - **Load Testing:** Simulate multiple users generating resumes or AI summaries simultaneously and measure system responsiveness.
 - **Stress Testing:** Test the system under extreme conditions (e.g., bulk AI requests) to identify the breaking point.
 - **Response Times:** Measure the time taken for:
 - Resume download
 - AI content generation
 - Resume preview loading
 - **Database Performance:** Evaluate the handling of bulk user resumes, templates, and AI-generated data.

5. Functional Testing

Functional testing ensures that the application features perform according to the requirements.

- **Tools:** Manual + Automated scripts
- **What to Test:**
 - User registration/login/logout
 - Resume creation (basic info, education, experience, skills, links)
 - AI Summary generation
 - Resume preview
 - Resume download (PDF)

6. UI/UX Testing

UI/UX testing checks the design consistency, responsiveness, and usability across various devices and browsers.

- **Tools:** BrowserStack, Chrome DevTools (Responsive mode), manual testing
- **What to Test:**
 - Layout responsiveness on mobile, tablet, and desktop
 - Readability and alignment of resume templates
 - Smooth navigation through dashboard and resume builder
 - Consistent design for form fields, buttons, and icons

7. Security Testing

Security testing helps ensure data protection and application safety.

- **Tools:** OWASP ZAP, manual checks, Burp Suite (optional)
- **What to Test:**
 - Password encryption and secure authentication
 - Input validation to prevent XSS and SQL injection
 - Secure access to user resumes (authorization checks)
 - Session expiration and logout mechanisms

8. Regression Testing

Regression testing ensures new features or fixes do not break existing functionality.

- **Tools:** Manual + Automation (Cypress/Jest)
- **What to Test:**
 - Re-run previous test cases after every update
 - Check critical user flows (login → create resume → download)
 - Confirm existing templates and AI features still function properly

9. User Acceptance Testing (UAT)

UAT validates whether the app meets user expectations in real-world usage.

- **Tools:** Feedback forms, Google Forms, or direct test sessions
- **What to Test:**
 - Overall experience of creating and downloading a resume
 - AI-generated summaries accuracy and usefulness
 - Ease of use for non-technical users
 - Navigation flow, template clarity, and personalization options

Test Cases

Test Case ID	Description	Precondition	Test Steps	Expected Result
tc_101	User Registration	User is on the signup page	1. Enter name, email, password 2. Click on “Sign Up”	User is registered and redirected to login page
Tc_102	User Login	User is registered	1. Enter email and correct password 2. Click on “Login”	User is logged in and redirected to dashboard
Tc_103	AI Summary Generation	User is logged in and on resume form page	1. Enter job role and job description 2. Click “Generate Summary”	AI-generated summary appears in the textbox
Tc_104	Resume Form Submission	User is logged in	1. Fill in all resume form fields (name, skills, experience, education) 2. Click “Submit”	Resume is saved and preview is generated
Tc_105	Resume Download as PDF	Resume is generated	1. Click “Download PDF” button	PDF file of the resume is downloaded
Tc_106	Logout Function	User is logged in	1. Click “Logout” button	User session is ended and redirected to login page

4.3 Integration System

An integration system in the AI Resume Builder ensures seamless interaction between various internal modules and external services like AI APIs, cloud storage, and resume templates. Proper integration ensures smooth data flow, optimal user experience, and automation of tasks such as AI summary generation and PDF export.

1. Internal Integration

1.1 User Management Integration

The user management module handles registration, login, and authentication for all users.

- **Resume Builder Access:** Once authenticated, users can access their dashboard, manage resume data, generate summaries, and download resumes.
- **Role-Based Control :** Admins can manage template access, view user stats, or moderate AI feedback logs.
- **Session Persistence:** Integrated with local/session storage or token-based authentication (JWT/Clerk) to maintain login state across pages.

1.2 Resume Module Integration

This is the core of the AI Resume Builder, enabling users to input their details and generate a resume.

- **AI Summary Integration:** The resume form connects with the AI API to generate job-role-specific professional summaries.
- **Form Autosave:** Integrated with local storage or backend API to autosave progress, ensuring data isn't lost during session expiry or accidental refresh.
- **Download Service:** Resume content is passed to a PDF generation engine which handles formatting and file export.

2. External Integration

2.1 AI APIs (e.g., OpenAI, HuggingFace)

- **Summary & Statement Generator:** Integrated with external AI services to process user input (job role, description) and return a professional summary.
- **Validation:** Handles API errors gracefully and displays fallback messages if the service is unavailable.
- **Token Usage Logging:** If billing or usage tracking is required, logs the number of requests made per user.

2.2 PDF Export & Cloud Storage

- **PDF Generation Library:** Integrated with libraries like html2pdf, jsPDF, or cloud-based solutions for exporting resumes to high-quality PDF formats.
- **Cloud Storage:** Integrated with services like AWS S3, Firebase, or Google Drive API to allow users to store and access resumes online.

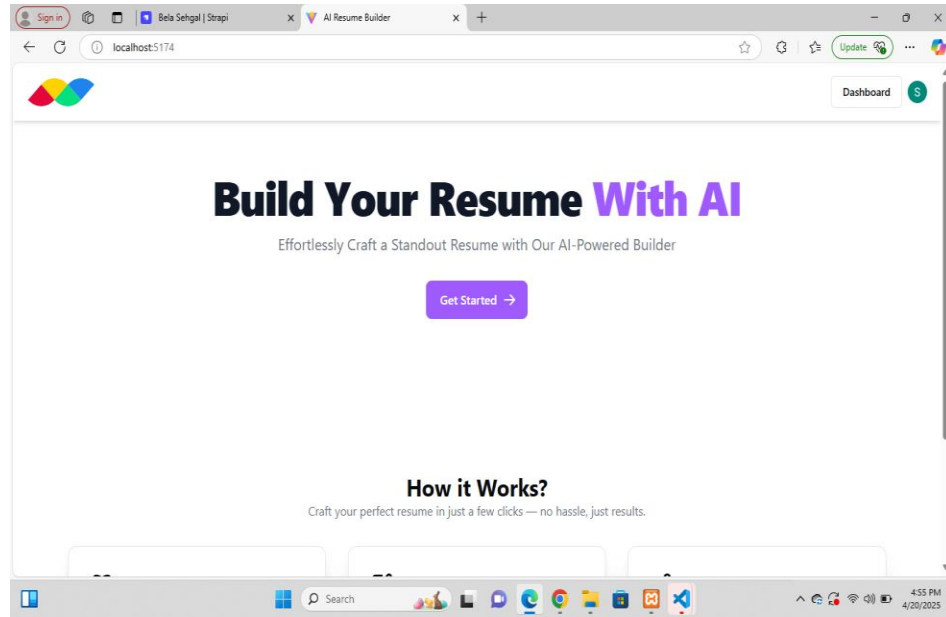
2.3 Resume Template Management

- **Dynamic Template Loading:** Integrated with a CMS like Strapi or a frontend store to fetch and display available template.
- **Version Control:** Ensures users always get the latest version of templates or styles without breaking their saved content.

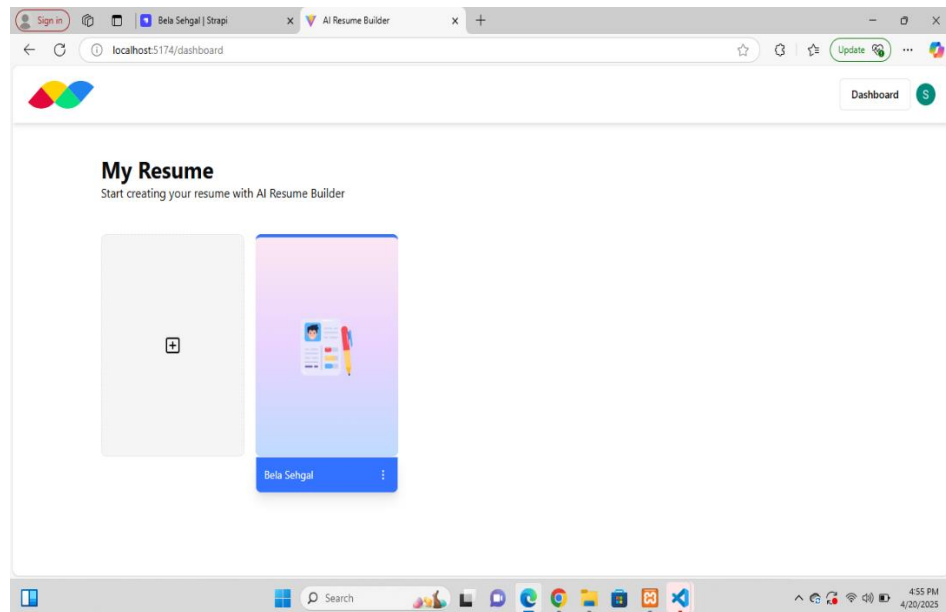
Chapter 5

Result

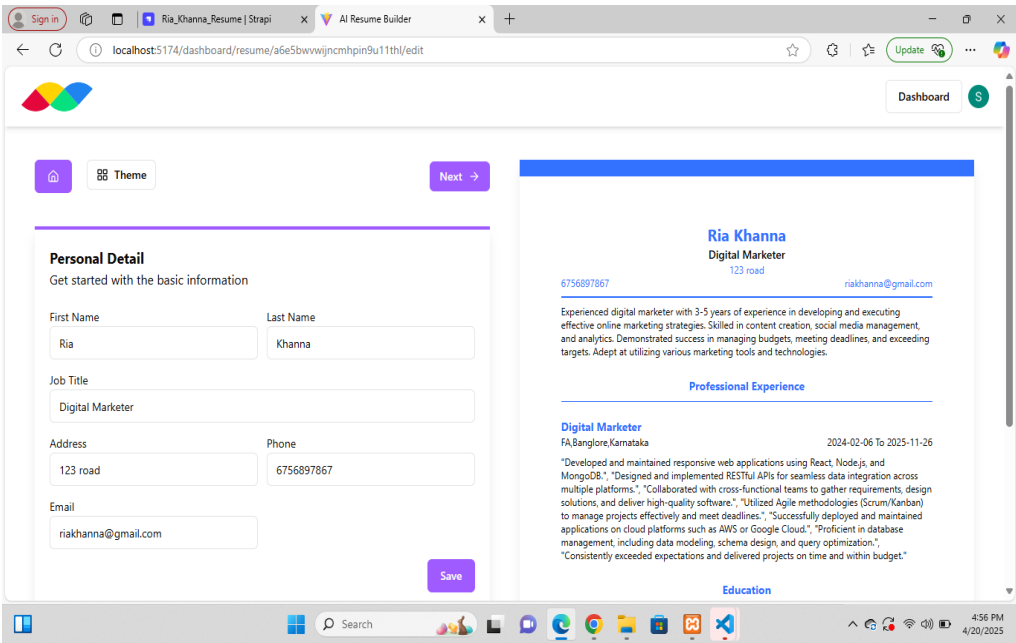
Homepage:



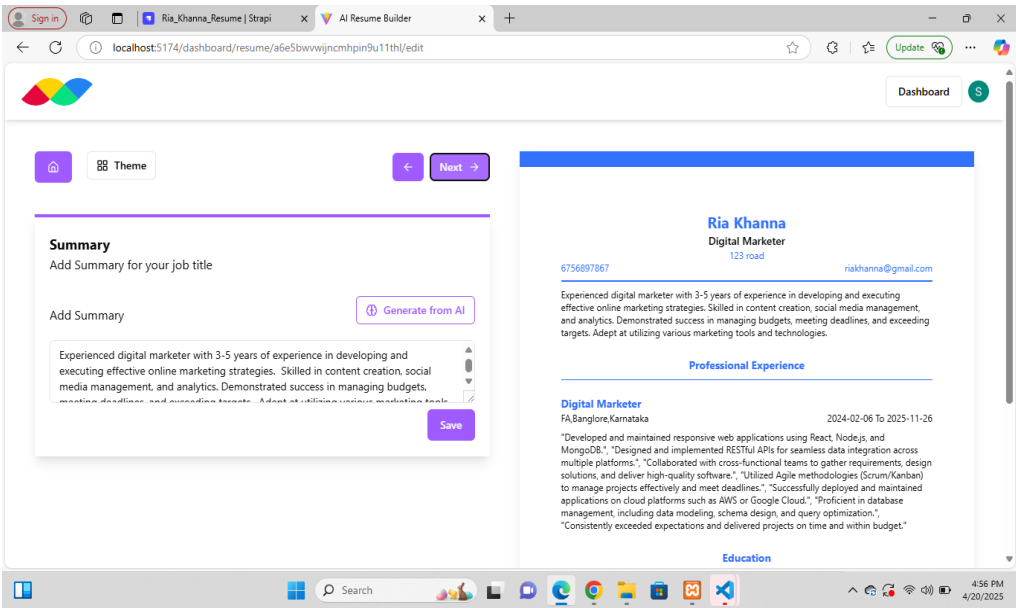
User Dashboard:



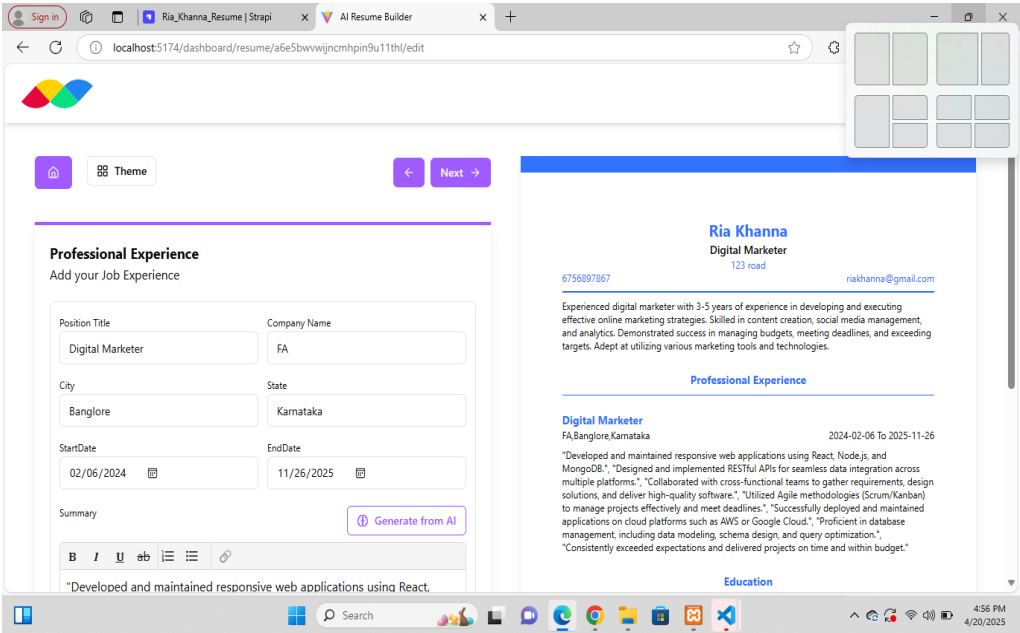
Personal Detail with Preview:



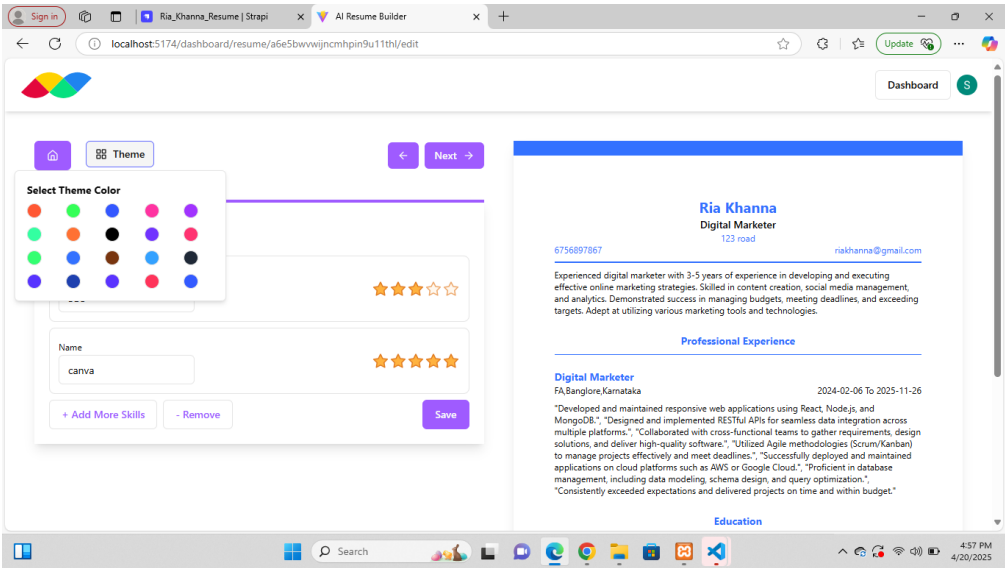
Summary with Preview:



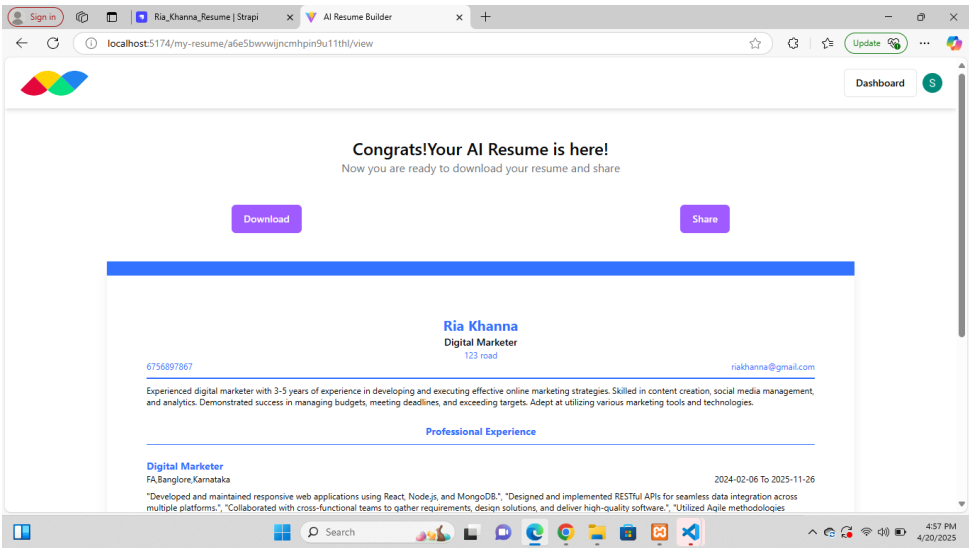
Professional Experience with Preview:



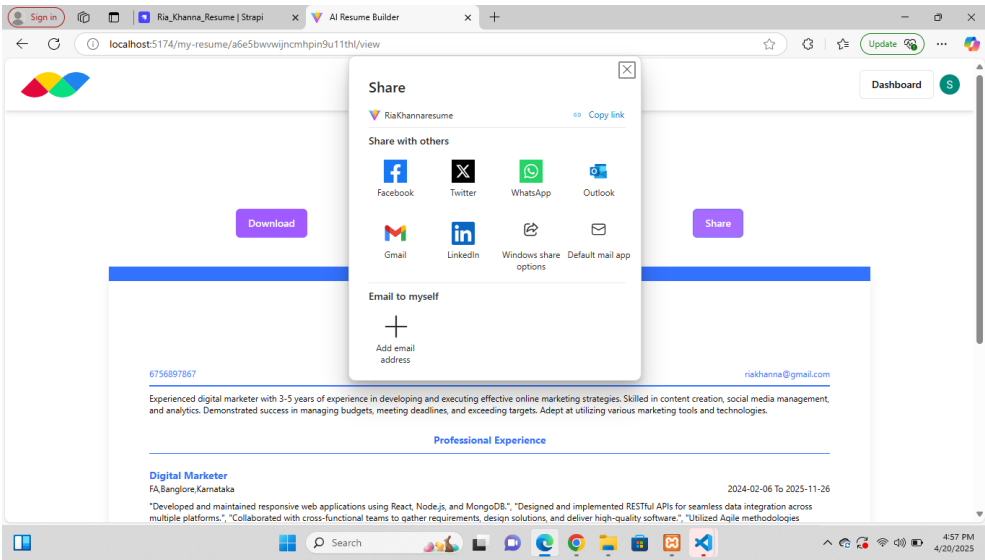
Skills and Theme Color:



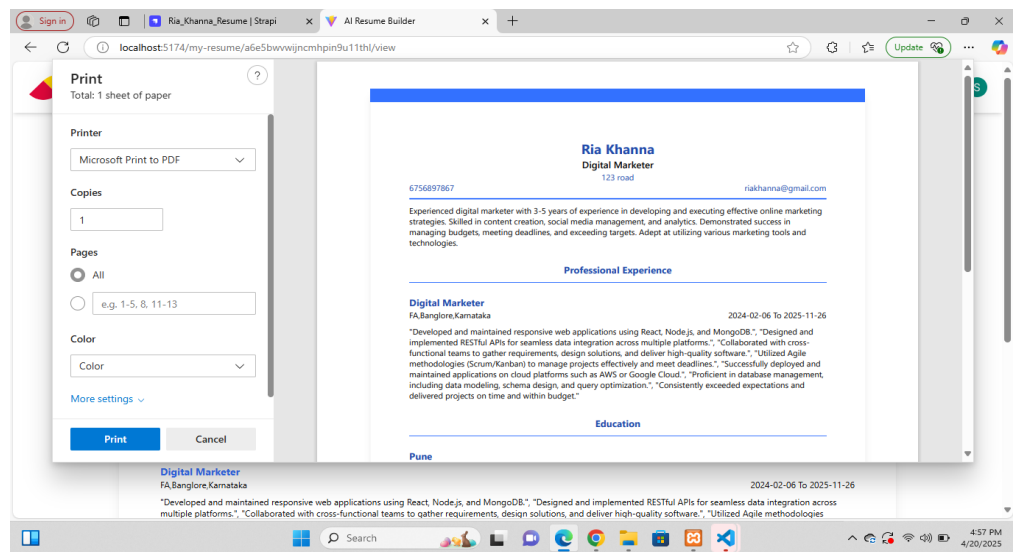
Resume:



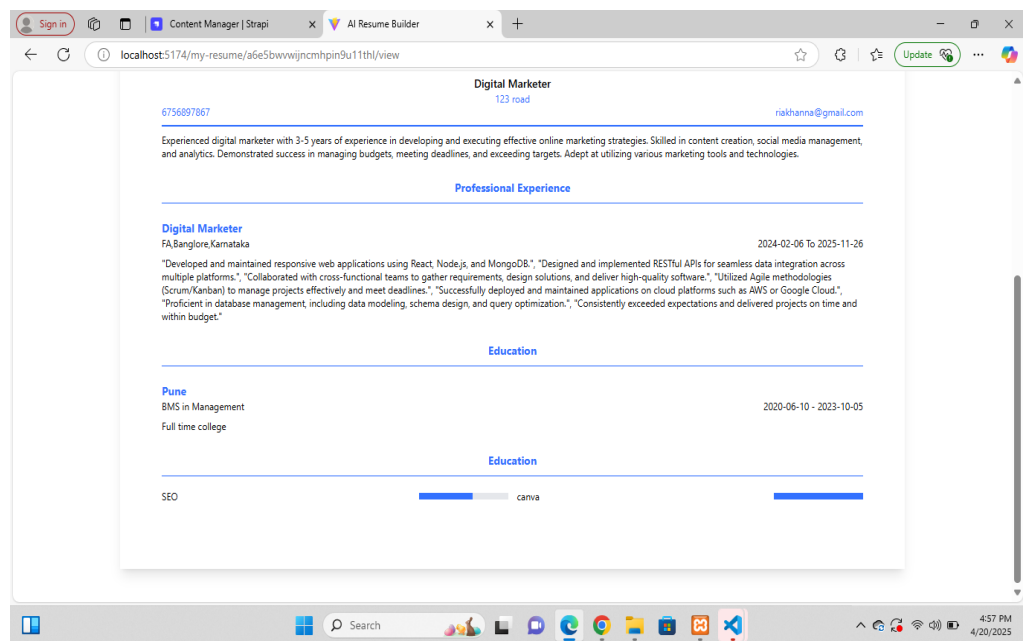
Share Resume:



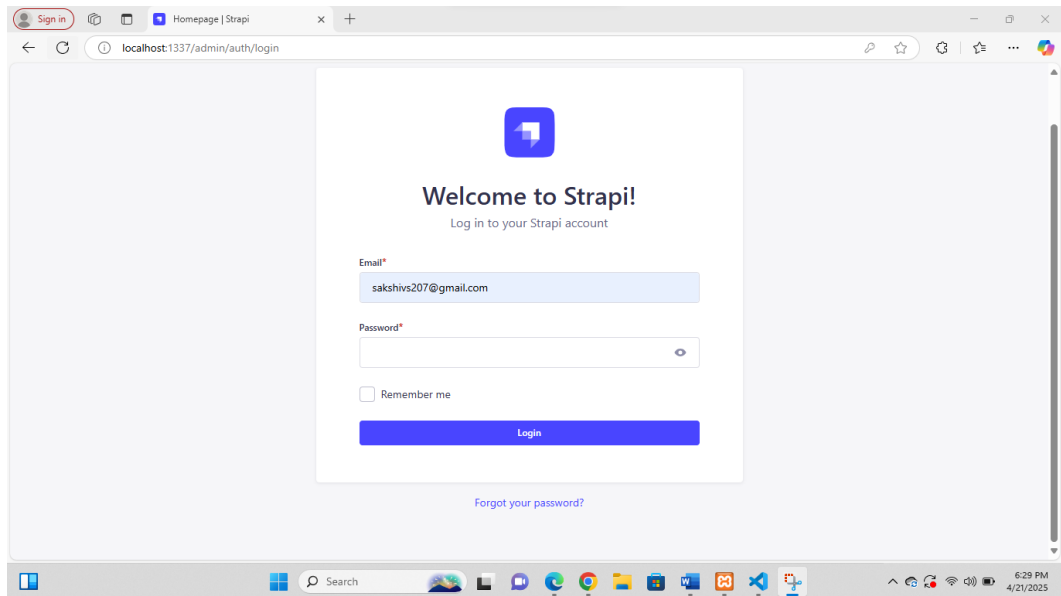
Download Resume:



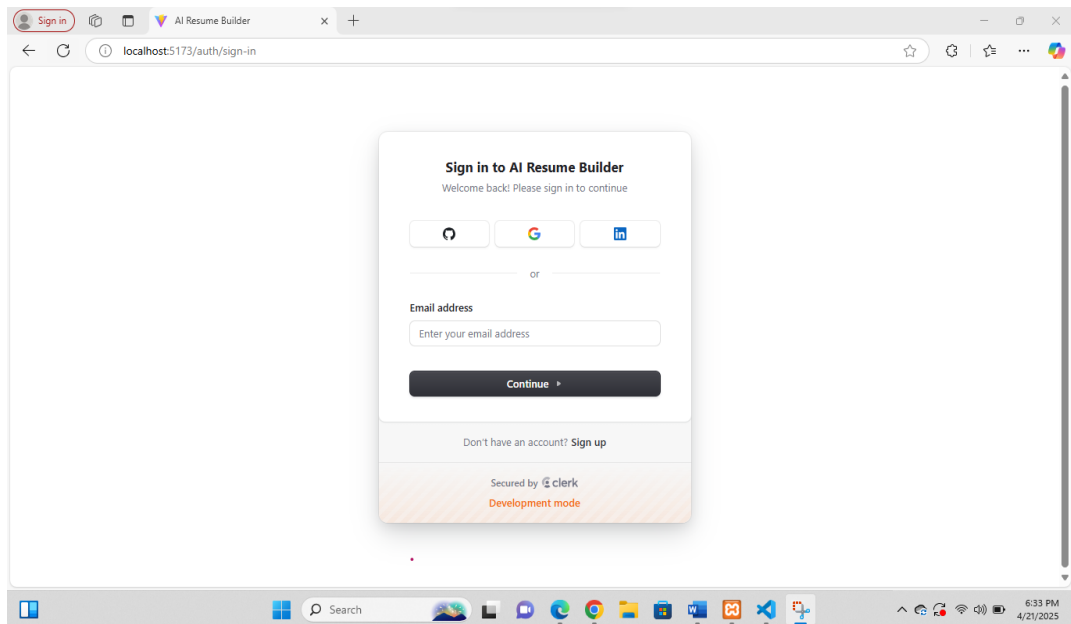
View Resume:



Admin Login:



User Login:



Chapter 6

Future Enhancements

While the current version of the AI Resume Builder offers essential functionality like user authentication, AI-powered summary generation, resume form filling, and PDF export, there is significant scope for improvement and expansion. The following are proposed future enhancements that can elevate the application's utility, performance, and user experience

1. Advanced AI Integration

- Implement domain-specific AI models to generate more tailored summaries for various industries (e.g., IT, Finance, Healthcare).
- Add functionality to generate customized cover letters using AI.

2. Template Marketplace

- Offer a broader selection of resume templates with customizable layouts and color schemes.
- Enable user-generated templates, allowing individuals to upload and share their own designs.

4. Enhanced Security and Privacy

- Implement data encryption for storing and transmitting personal information.
- Introduce two-factor authentication for added login security.
- Include user consent forms for data processing, aligning with privacy regulations.

5. Interactive Assistant

- Introduce a virtual assistant or chatbot to guide users through each section of the resume builder.
- Provide real-time tips and suggestions while the user is filling out information.

Chapter 6

Conclusion

The AI Resume Builder project was developed with the aim of simplifying the resume creation process for job seekers by leveraging modern web technologies and artificial intelligence. The system provides an intuitive platform where users can enter their personal and professional details, generate AI-powered summaries, select from pre-designed templates, and export high-quality resumes in PDF format.

Through the integration of AI, the builder enhances the quality and effectiveness of resumes by tailoring content to job roles and improving the overall presentation. The system also ensures data security, usability, and scalability, making it suitable for a wide range of users from fresh graduates to experienced professionals.

This project not only demonstrates the potential of combining AI with resume-building tools but also opens up various possibilities for future enhancements, such as advanced analytics, cloud integration, job portal connections, and multilingual support.

In conclusion, the AI Resume Builder is a valuable solution in today's competitive job market, helping individuals create professional resumes quickly and efficiently with the assistance of intelligent automation.

Chapter 8

References

1.	https://www.google.com
2.	https://tailwindcss.com